

Most economic observers in the Republic of Nearland agree that the country will suffer a recession in the near future. At an economics conference at Nearland's premier university, four professors were having a lively discussion about how economic theory, the business cycle, and investment performance interact.

Professor Adams made two statements about the utility of consumption being an important driver of interest rates:

Statement 1: "If we believe that a recession is likely in the future, we would expect the marginal utility of consumption in the future to be lower relative to the utility of current consumption, and the inter-temporal rate of substitution to be higher than they would otherwise be if there was optimism about future economic conditions."

Statement 2: "It is my estimate that consumption in one year's time has 5% less utility than consumption in the present."

Professor Brady poured scorn on Professor Adams' second statement. He replied that predicting such precise values for abstract economic concepts such as utility was impossible. Trying to value bonds and interest rates, he argued, was much more likely to be accurate if calculations were based on more measurable macroeconomic fundamentals, such as GDP growth and inflation. Central bank policy rates, he stated, are positively correlated to current inflation and current GDP growth.

Professor Chapman attempted to mediate between Adams and Brady. He said, "In a way, you both have a point. At the top of the business cycle there is higher inflation and current GDP growth. At the same time, market participants begin to worry about future recession, which increases the marginal utility of delayed consumption. These conditions both explain a steepening of the upward-sloping yield curve."

Professor Douglas tried to move the conversation towards the business cycle and risk premiums. She presented the audience with economic data as presented in Exhibit 1, and challenged observers to calculate the equity risk premium in Nearland.

Exhibit 1: Economic Data for Nearland

Break-even rate of inflation	2.80%
Credit spread	3.10%

Risk premium for equities relative to debt 4.50%

Professor Douglas went on to discuss asset classes, which could act as a consumption hedge to guard against the upcoming recession. She invited the audience to attend her upcoming discussion seminar where she would analyze the prospects for real estate, growth stocks, and value stocks.

Question #1 - 6 of 31

Question ID: 1474026

Statement 1 made by Professor Adams is *most* likely to be:

- A) correct.
 - B) incorrect in respect of the marginal utility of consumption.
 - C) incorrect in respect of the inter-temporal rate of substitution.
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Question #2 - 6 of 31

Question ID: 1474027

If Statement 2 made by Professor Adams is correct, the one year real risk-free rate of return will *most* likely be closest to:

- A) 4.95%.
 - B) 5.00%.
 - C) 5.26%.
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Question #3 - 6 of 31

Question ID: 1474028

Assuming the Taylor rule holds, Professor Brady's statement on the correlation of interest rates with macroeconomic fundamentals is *most* likely:

- A) correct.
 - B) incorrect in respect of the correlation of interest rates with inflation.
 - C) incorrect in respect of the correlation of interest rates with GDP growth.
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Question #4 - 6 of 31

Question ID: 1474029

Professor Chapman's statement about utility, inflation, GDP growth, and the yield curve is *most* likely:

- A) correct.
 - B) incorrect as the circumstances described would have no effect on the yield curve.
 - C) incorrect as the circumstances described would flatten an upward-sloping yield curve.
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Question #5 - 6 of 31

Question ID: 1474030

The equity risk premium in Nearland is *closest* to:

- A) 4.50%.
 - B) 7.60%.
 - C) 10.40%.
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Question #6 - 6 of 31

Question ID: 1474031

Of the asset classes mentioned by Professor Douglas, the *most* likely to be suitable as a consumption hedge is:

- A) real estate.
 - B) growth stocks.
 - C) value stocks.
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Question #7 of 31

Question ID: 1474010

Janet Grange's current one-period inter-temporal rate of substitution is 0.95. Janet is *most likely* to invest in a default-free inflation indexed one-period bond if:

- A) The bond's return is 4.89% or more.

B) The bond's return is 5.26% or more.

C) The bond's return is 5% or more.

Question #8 of 31

Question ID: 1474021

ABC Inc. stock's price is inversely related to the business cycle; it is higher during economic downturns. Which of the following appropriately characterizes the consumption hedging property of an investment in ABC stock and the equity risk premium demanded by investors for an investment in it?

A) Due to its desirable consumption hedging ability, an investment in ABC stock would command a higher equity risk premium.

B) Due to its desirable consumption hedging ability, an investment in ABC stock would command a lower equity risk premium.

C) Due to its poor consumption hedging ability, an investment in ABC stock would command a higher equity risk premium.

Question #9 of 31

Question ID: 1474032

As compared to an investment in equities, the difference in discount rate for valuation of commercial real estate is *most likely* due to:

A) the break-even inflation rate.

B) inflation uncertainty.

C) lack of liquidity.

Question #10 of 31

Question ID: 1474011

The risk-premium for uncertainty in inflation is *most likely* to be:

A) unrelated to the term spread.

B) negatively related to the term spread.

C) positively related to the term spread.

Question #11 of 31

Question ID: 1474015

Credit spreads are *most likely* to:

- A) Rise during expansions.
 - B) Rise during recessions.
 - C) Rise with policy rates.
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Question #12 of 31

Question ID: 1474014

Break-even inflation rate is *most appropriately* described as the:

- A) The difference between market's expectation of the inflation rate and the risk premium for inflation uncertainty.
 - B) The difference in yields between long-dated and short-dated government bonds.
 - C) The difference in yields of non-inflation indexed and inflation indexed risk-free bonds.
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Question #13 of 31

Question ID: 1474003

Market values of assets are *most likely* to be affected when:

- A) New information reveals that the market's expectations about earnings were accurate.
 - B) New information reveals that the market's expectations about earnings were inaccurate.
 - C) New information confirms the market's expectations of future earnings.
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Question #14 of 31

Corporate earnings from which of the following sectors would be most sensitive to business cycle?

- A) Consumer durable
 - B) Consumer non-discretionary
 - C) non-cyclical
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Question #15 of 31

Question ID: 1474005

The marginal utility of current consumption is *most likely* higher:

- A) During economic contractions.
 - B) During economic expansions.
 - C) Higher the wealth.
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Question #16 of 31

Question ID: 1474019

Spreads for issuers in the consumer cyclical sector are *most likely* to:

- A) Be unrelated to business cycle.
 - B) Decrease during economic downturns.
 - C) Increase during economic downturns.
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Question #17 of 31

Question ID: 1474012

Which of the following is *most likely* to be the shape of the yield curve during recessions?

- A) Upward sloping
 - B) Downward sloping
 - C) Flat
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Question #18 of 31

Question ID: 1474009

The price of a zero-coupon, inflation indexed, risk-free bond that pays \$1 in one period is:

- A) Uncertain.
 - B) The expected value of the investors' inter-temporal rate of substitution between current period and one period from now.
 - C) \$1.00
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Question #19 of 31

Question ID: 1474023

Price multiples are *least likely* to increase when:

- A) Inflation expectations decline.
 - B) Equity risk premium increases.
 - C) Earnings growth increases.
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Question #20 of 31

Question ID: 1474013

If the market expects inflation to decrease over the next few years but the uncertainty about inflation was increasing, the break-even inflation rate is *most likely* to:

- A) Increase.
 - B) Be uncertain.
 - C) Decrease.
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Question #21 of 31

Question ID: 1474002

Market values of assets are *most likely* to be affected when either:

- A) Risk free interest rates, risk premiums, timing and/or magnitude of expected cash flows change.

- B) Real risk-free rates, risk premiums, timing/magnitude of expected cash flows change.
 - C) Real risk-free rates, inflation premium, timing/magnitude of expected cash flows change.
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Question #22 of 31

Question ID: 1474017

Jeff Dentmat is expecting overall credit spreads to narrow over the next few years. Which of the following conclusions can Dentmat most appropriately make?

- A) Government bonds will outperform low-rated corporate bonds.
 - B) Risky bonds will outperform risk-free bonds.
 - C) High rated corporate bonds will outperform low-rated corporate bonds.
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Question #23 of 31

Question ID: 1474007

A high default-free interest rate is *most likely* to be associated with:

- A) expectations of higher income in the future.
 - B) investors attaching high utility to future consumption relative to current consumption.
 - C) expectations of goods and services being less available in the future relative to today.
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Question #24 of 31

Question ID: 1474018

Differences in credit spreads across sectors are *least likely* due to:

- A) Differences in leverage typical of the sector.
 - B) Differences in ratings.
 - C) Differences in services and products that an industry produces.
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Question #25 of 31

Question ID: 1474020

Which of the following assets provides a most effective hedge against bad consumption outcomes?

- A) Equity
 - B) Risk-free bonds.
 - C) Real estate
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Question #26 of 31

Question ID: 1474001

Which of the following is *least likely* to explain a decline in the S&P 500 index:

- A) A decrease in expectations about corporate earnings.
 - B) An increase in Treasury yields.
 - C) A decline in expected inflation.
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Question #27 of 31

Question ID: 1474006

Rapidly developing economies like India and China have high GDP growth rates and therefore are *most likely* to have a:

- A) Low real rate, high inter-temporal rate of substitution and a low rate of current borrowing by investors.
 - B) High real rate, low inter-temporal rate of substitution and a high rate of current borrowing by investors.
 - C) High real rate, low inter-temporal rate of substitution and a low rate of current consumption.
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Question #28 of 31

Question ID: 1474008

The real rate of return is *most likely* higher:

- A) Higher the utility investors attach to future consumption relative to current consumption.
- B) Higher the expectations of lower income in the future.
- C) Lower the utility investors attach to future consumption relative to current consumption.
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Question #29 of 31

Question ID: 1474004

Investors are *least likely* to increase their savings rate when:

- A) Uncertainty about their future income decreases.
- B) Uncertainty about their future income increases.
- C) Expected rates of returns increase.
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Question #30 of 31

Question ID: 1474024

Value stocks are *most likely* to be characterized by:

- A) Immature markets.
- B) High earnings growth.
- C) Low price multiples.
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Question #31 of 31

Question ID: 1474016

The break-even inflation rate is expected to be 2% over the next year. What is the credit spread for a 2% annual pay corporate bond maturing in one year with a market price of \$96.91 (\$100 par) if the real risk-free rate of return over the next year is 1%?

- A) 0.19%
- B) 2.00%
- C) 2.25%